

EXPERIMENT - 8

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8.1.2. Prime Numbers in a Given Range

A) ALGORITHM

Step 1: Start

Step 2: Read the starting number (start).

Step 3: Read the ending number (stop).

Step 4: For num = start to stop

 If num > 1

 Set flag = True

 For i = 2 to √num

 If num % i == 0

 Set flag = False

 Break

 End For

 If flag == True

 Print num

 End If

Step 5: Stop

B) PYTHON CODE

```
start = int(input())
```

```
stop = int(input())
```

```
f=False
```

```
for num in range(start, stop + 1):
```

```
    if num > 1:
```

```
        flag = True
```

```
    for i in range(2, int(num ** 0.5) + 1):
```

```
        if num % i == 0:
```

```
            flag = False
```

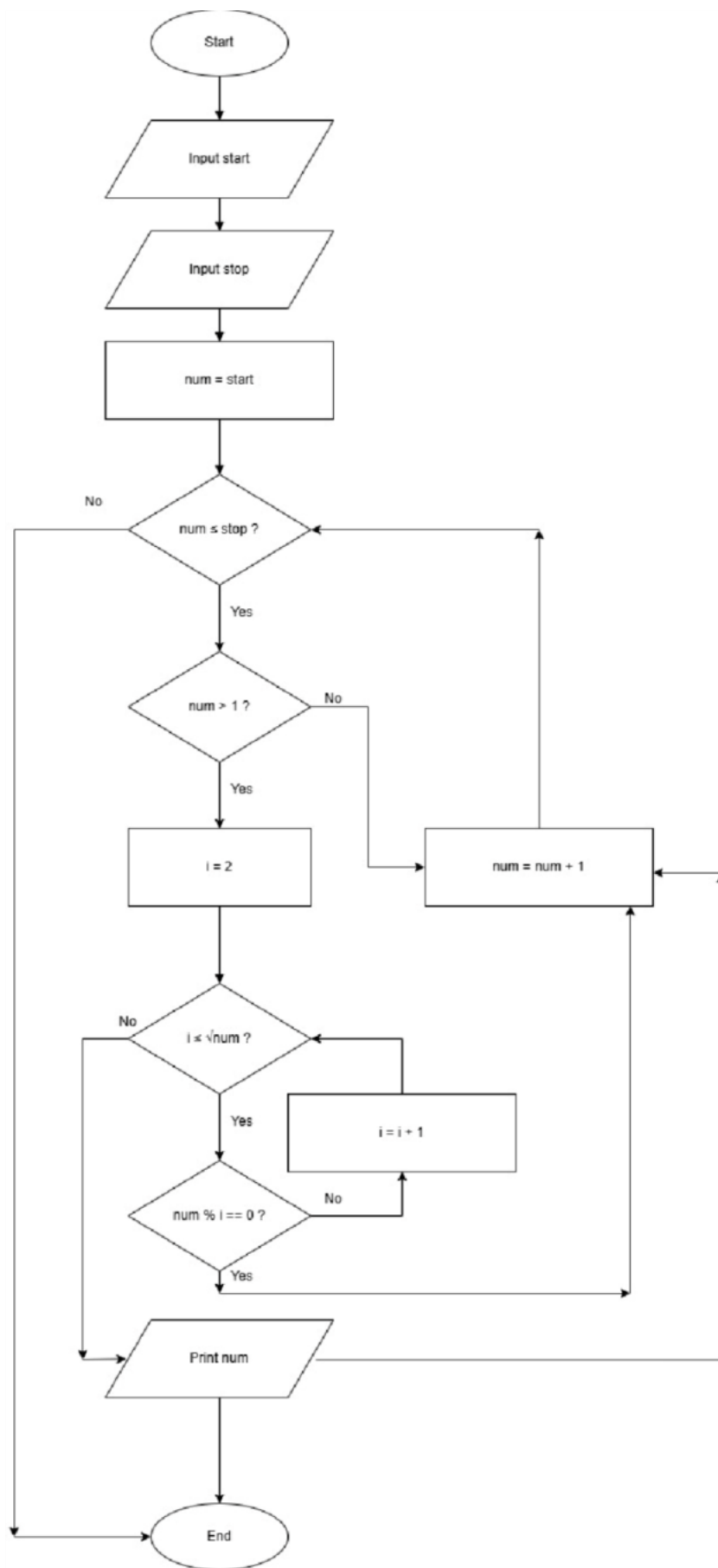
```
            break
```

```
    if flag
```

```
        print(num)        f=True
```

```
if not f:print("No primes")
```

C) FLOWCHART



8.1.2. Prime Numbers in a Given Range

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases

primeNu...
Submit

```

1 start = int(input())
2 stop = int(input())
3 f=False
4 for num in range(start, stop + 1):
5     if num > 1:
6         flag = True
7
8     for i in range(2, int(num ** 0.5) + 1):
9         if num % i == 0:
10            flag = False
11        break

```

Average time
0.010 s
9.66 ms

Maximum time
0.014 s
14.00 ms

Test case 1 **14 ms** Debug

Expected output	Actual output
10	10
20	20
11	11
13	13
17	17
19	19

5 out of 5 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Terminal Test cases

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